

THE CAPTURE THEORY AND PLANETARY CONDENSATION

J. R. Dormand and M. M. Woolfson

(Received 1970 October 12)

SUMMARY

The condensation of planets under the conditions of Woolfson's capture theory of the origin of the solar system is described. It is shown that material captured by the Sun, during an encounter with another star, would probably not recondense to form planets under the conditions originally specified. Different star encounters are studied using numerical analysis, and it is shown that as a result of a 'fast' encounter, the Sun could capture material which would recondense to form planets and take up orbits which are comparable to those observed in the solar system.

The probability of such star encounters, and also the origin of planetary satellites, are briefly discussed.

1. INTRODUCTION

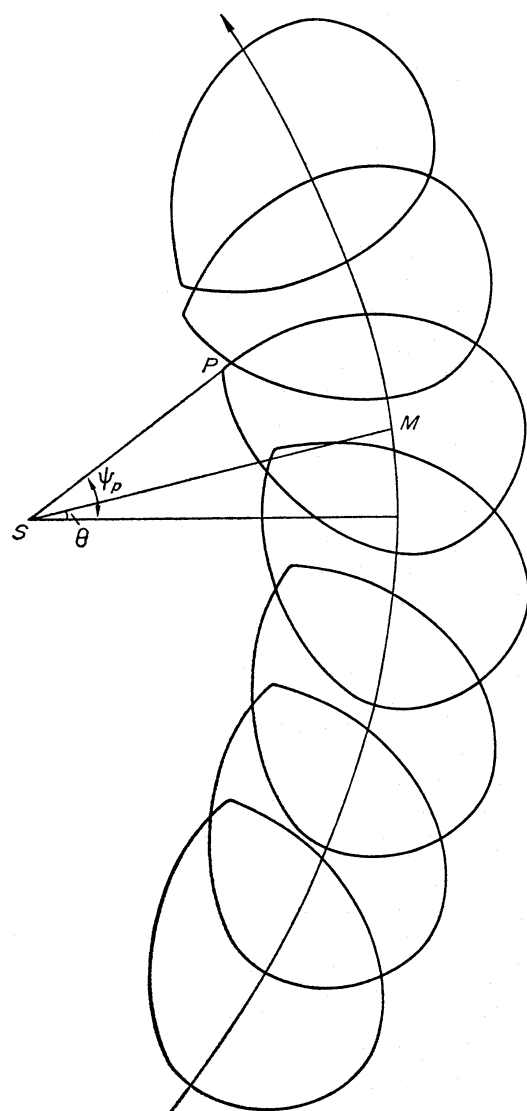
In 1964 a capture theory of the origin of the solar system was put forward by Woolfson. It was proposed that the Sun could have captured planetary material from a light, diffuse star by tidal forces during an encounter, and it was shown that for some set of parameters the captured material could take up a range of orbits around the Sun with perihelia comparable to those observed in the solar system. The elements of these orbits are shown in Fig. 1; they have been corrected since the detection of an error in the original computer program, but they are not significantly different from those originally published (Woolfson 1964). The figure of the passing star was discussed in the earlier work and is also considered in Section 5 of the present paper.

The condensation of planets from the captured material was investigated according to the theory of the formation of condensations in a gaseous filament, as given by Jeans (1919), and it was shown that the ejected filament may have been sufficiently dense to have given rise to planets.

In the present paper an investigation into the evolution of the planetary material, in a non-uniform, time-varying gravitational field gradient, will be described. It will be shown that the condensation of planets is unlikely under the conditions originally specified. A different method for the simulation of star encounters by numerical analysis will be described, and this will be used to examine the possibility of the formation of the solar system as a result of a 'fast' encounter between the Sun and another star.

2. CONDENSATION OF A PLANETARY FILAMENT

No attempt was made in the 1964 paper to consider the effect of external gravitational fields on the evolution of the captured planetary filament. Since a star has been tidally disrupted by the effect of the Sun, and the ejected material is still under the



Time of ejection (10^8 s)	Perihelion (A.U.)	Eccentricity
0	31.0	0.043
4	25.7	0.083
6	23.0	0.136
8	22.0	0.163
10	21.0	0.188
12	18.4	0.255
14	14.7	0.358
16	10.3	0.505
18	5.4	0.707
20	1.7	0.898
22	0.13	0.992
24	0.05	0.997

FIG. 1. A succession of configurations of a distorted star as it moves past the Sun. Intervals of 4×10^8 s. The Table gives perihelion distances and eccentricities of material escaping from the tidal extremity at various times.

influence of the Sun's field gradient, it would seem unlikely at first glance that it could condense to form planets. However, consideration of the evolution of the filament must also take into account the gravitational field of the receding star. Clearly, if the Sun, star, and filament are collinear, the tidal effects would be a maximum and the filament would be drawn out into a narrow stream. One of the most widely quoted objections to the early tidal theories of Jeffreys (1918) and Jeans (1919), was put forward by Spitzer (1939). He suggested that hot material ejected

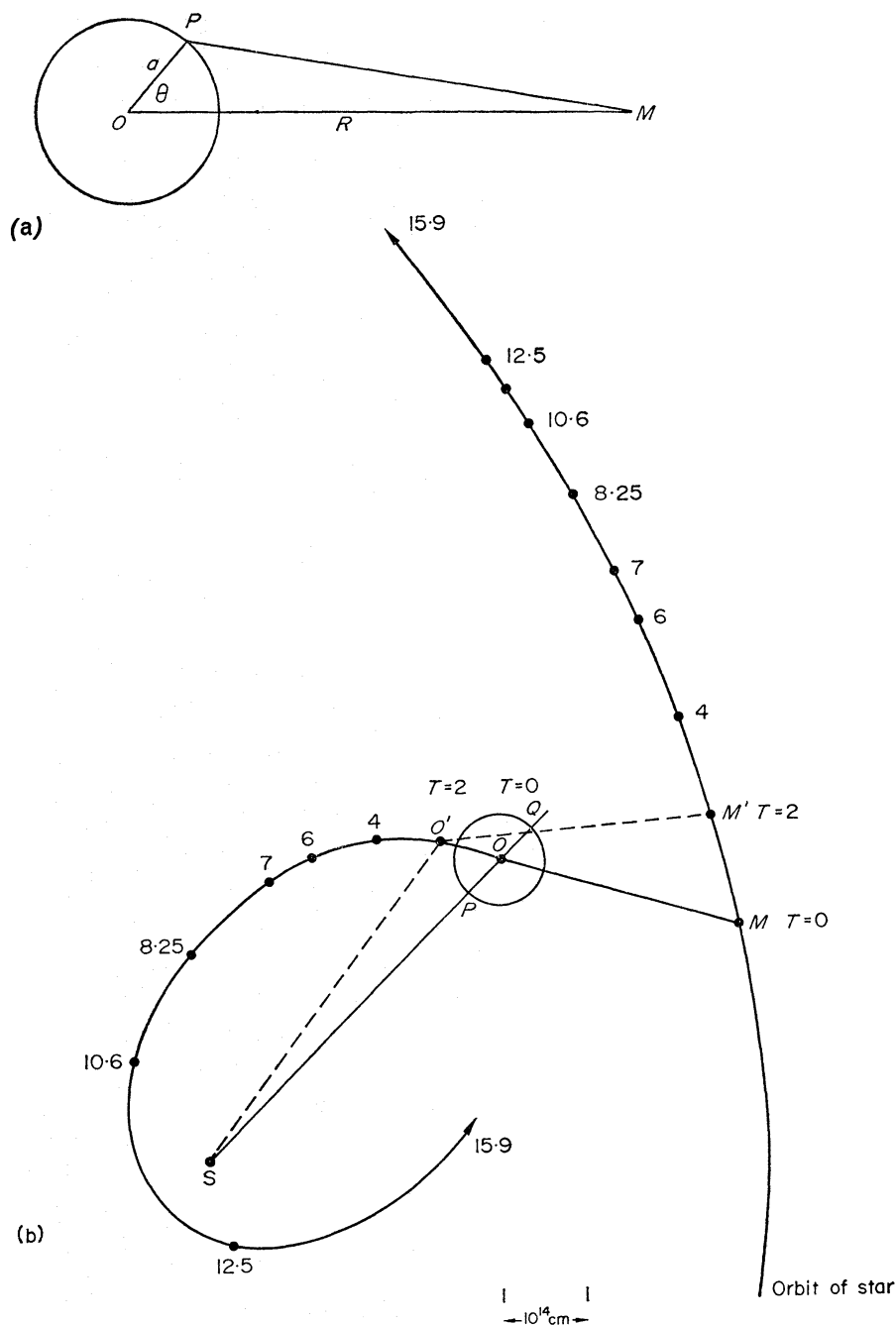


FIG. 2. (a) Geometry of a proto-planet, centre O , and tide-producing mass at M .

(b). Orbit of star and a proto-planet captured by Sun (S) at time 1.8×10^9 s. Mass and perihelion of star are 3×10^{32} g and 6.67×10^{14} cm, and times are given at intervals on each orbit.

from the Sun would have dissipated violently into space within a matter of minutes, rather than have condensed into planets. However Lyttleton (1960) has pointed out that the filament would be restrained from expanding into space by the gravitational field gradient, perpendicular to the line of the filament, due to the two separating stars which would not have separated appreciably in a few hours. In the case of the capture theory being discussed here, the temperature of the outer regions of the star from which the filament was drawn, will probably be about 50°K rather than a million degrees as in the case studied by Spitzer, and so the thermal speeds of the gas molecules will be less by two orders of magnitude.

In some circumstances the field gradients of the Sun and star may, to some extent, counteract each other. The most favourable situation exists when the two fields are perpendicular to each other. Fig. 2(b) shows the orbit of material ejected from the star at $T = 1.8 \times 10^9$ s (see Fig. 1). It can be seen that angle SOM is approximately 120° , and so the planetary material will, in this case, be partially 'sheltered' from the tidal forces due to the Sun. It is possible to make an estimate of the magnitude of this 'sheltering' effect in this particular situation. The tidal potential at a point P (see Fig. 2(a)) of a body, due to a mass M , can be written

$$V_t = \frac{-GM}{R} \left\{ \left(\frac{a}{R}\right)^2 P_2(\cos \theta) + \left(\frac{a}{R}\right)^3 P_3(\cos \theta) + \left(\frac{a}{R}\right)^4 P_4(\cos \theta) + \dots \right\}$$

Differentiating with respect to 'a' we obtain the radial acceleration along OP due to this tidal potential, i.e.

$$\ddot{r} = \frac{GMa}{R^3} \left\{ 2P_2 + 3\left(\frac{a}{R}\right)P_3 + 4\left(\frac{a}{R}\right)^2 P_4 + \dots \right\}$$

where P_2 , P_3 and P_4 are Legendre polynomials. Consider now a proto-planet at the time of its ejection from the star (Fig. 2(b)). Using the above formula, we find that the tidal accelerations along OP and OQ , due to the Sun, are 1.25×10^{-4} and $0.93 \times 10^{-4} \text{ cm s}^{-2}$ respectively, whilst those due to the star, in the same directions, are -2.34×10^{-6} and $-1.97 \times 10^{-5} \text{ cm s}^{-2}$. Thus we see that the proto-planet is afforded some degree of shelter, particularly on that side nearest to the star. As the star recedes, so its effect will lessen, and it was found that when angle SOM increases beyond 130° , the tidal effect of the star will reinforce that of the Sun. The 'sheltering' will therefore only occur for about 2×10^8 s after the ejection of the proto-planet; during this time the proto-planet moves from O to O' , and the star from M to M' (Fig. 2(b)). It seems therefore that the material will have to have condensed considerably soon after ejection from the star if it is to survive the large gravitational field gradient of the Sun at perihelion.

3. A DETAILED MODEL OF PLANETARY EVOLUTION

To test whether or not a planetary condensation could form in the stellar filament a model planet has been devised which is amenable to treatment by numerical analysis. The planet was represented by a number of discrete point masses distributed on a cubical lattice contained within a spherical volume. The initial density was taken to be uniform and the point masses were made equal. Differential equa-

tions of motion were set up in three directions for each mass point. The gravitational forces acting on each mass point included:

- (i) Force due to Sun;
- (ii) Force due to receding star which was taken to be a point mass on a parabolic orbit with respect to the Sun;
- (iii) Forces due to all other planetary mass points.

An unnatural feature of mass point models as used in this simulation, is the tendency for the points to 'pool' together in a number of distinct groups, thus giving the body a fragmented appearance (see Fig. 7, Woolfson 1964). To prevent this 'pooling', a repulsion force was introduced between the mass points, corresponding to the physical reality of gaseous matter resisting the tendency to build up regions of high density. This was similar to the 'elbow' factor used by Woolfson (1964) and was combined with the gravitational force between points so as to make the combined term positive or negative, depending on their separation. Suppose the force on point i due to a point j can be written

$$F_{ij} = -\frac{GM_iM_j}{r_{ij}^2} + \frac{\Psi_{ij}}{r_{ij}^3} \quad (1)$$

where M_j is the mass of the j th point, r_{ij} is the separation of the points, and Ψ_{ij} is a parameter depending on r_{ij} . Let the critical distance between the points be d_{ij} , so that we require attraction if $r_{ij} > d_{ij}$ and repulsion if $r_{ij} < d_{ij}$. If $r_{ij} = d_{ij}$, $F = 0$, so from (1) $\Psi_{ij} = GM_iM_j d_{ij}$. If we make d_{ij} proportional to the root mean square of the distances of two interacting points from the centre of the planet, i.e.

$$d_{ij} = \xi \left(\frac{s_i^2 + s_j^2}{2} \right)^{1/2}$$

where s_i is the distance of point i from the centre and ξ is constant, then the complete point interaction force can be written

$$F_{ij} = -\frac{GM_iM_j}{r_{ij}^2} \left[1 - \frac{\xi}{r_{ij}} \left(\frac{s_i^2 + s_j^2}{2} \right)^{1/2} \right].$$

This makes the critical separations between mass points proportional to their distances from the centre of the planet model, and allows condensation to take place without limit but prevents 'pooling'. Another effect is that condensation is most favoured in the central region, a characteristic that would be expected for a real planet. The value of ξ , the 'elbow' parameter, should clearly be in the range, 0 to 1, although values near unity will be unacceptable since, initially, adjacent pairs of points near the boundary of the model, ($s_i, s_j > r_{ij}$), will experience mutual repulsion forces, and contraction will be significantly retarded. If the initial radius of the model is n times the point separation, the maximum value of ξ , to ensure no repulsion in the basic point configuration, should be $1/n$. The best value of ξ was determined empirically as a result of several test runs of the computer program.

Another non-gravitational force has been added to the equations of motion to simulate the effect of viscous forces in the condensing proto-planet and also to control its rate of contraction. This force is such that the acceleration of a point is

$$\chi \dot{s}_i / s_i^2$$

where χ is constant and $\dot{s}_i = ds_i/dt$.

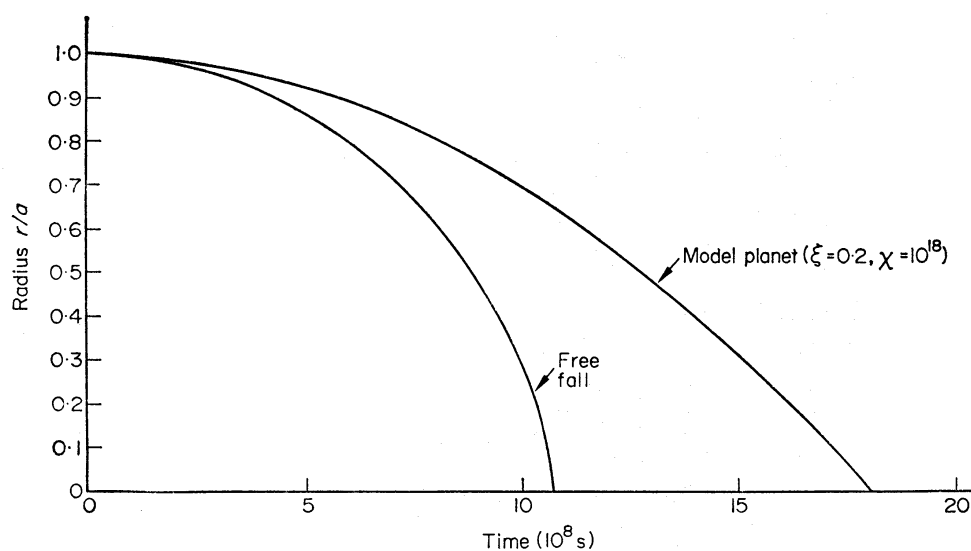


FIG. 3. Contraction of model planet compared with free fall contraction.

The equations of motion are integrated step by step using a four-stage Runge-Kutta method. The integration time-step length will depend on the mass-point separations, and therefore it must be recomputed after each step of the computation. An approximate integration of the one-dimensional equation of motion of two mass points shows that the step length should be proportional to $r^{3/2}$. In the planet model

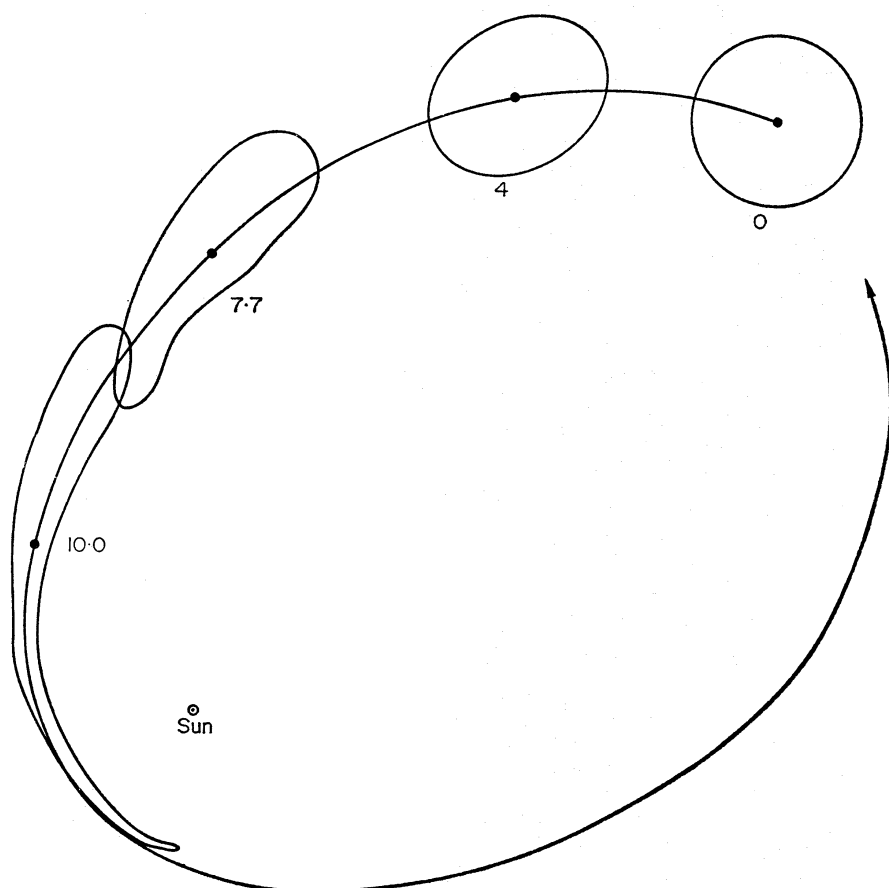


FIG. 4. Disruption of a model planet in the orbit shown in Fig. 2.

the value of r was taken to be the separation of the closest pair of points, and the initial time step was determined empirically.

A model planet, in which the initial point separation is one-third of the radius of the proto-planet was chosen as being sufficiently detailed for the simulation. This model has 123 mass points but, since the Sun, star and planet centre are taken to be coplanar, only one hemisphere needs to be treated independently; the other will remain a mirror image of this. The number of independent points is therefore 76, consisting of those coplanar with the Sun and star, and those in the upper hemisphere.

The first tests of the model were made with the planet in isolation so that its contraction characteristics could be determined. It was suggested by Woolfson (1964) that the behaviour of such a proto-planet may be similar to that of the cloud of gas and dust studied by Rouskol (1955). If this is so, we may expect appreciable contraction to have taken place in about twice the time required for free fall gravitational contraction. The viscosity constant ' χ ', which was included to control the rate of contraction, was chosen empirically with regard to this suggestion. The behaviour of the model was found to be not sensitively dependent on its value, provided it was of the correct order. In Fig. 3 the contraction curves for the model planet and a sphere of gas of the same density are shown. With an 'elbow' parameter, ξ , of 0.2 and viscosity constant, χ , equal to $10^{18} \text{ cm}^2 \text{ s}^{-1}$ the time of contraction of the model planet is about twice that of free fall contraction. The proto-planet was taken to have a mass of $2 \times 10^{30} \text{ g}$ and a radius $5 \times 10^{13} \text{ cm}$, giving a density of $4 \times 10^{-12} \text{ g cm}^{-3}$.

The model planet was next placed in orbit about the Sun at its supposed time of ejection from the star (see Figs 1 and 2). Fig. 4 shows the surface profile of the model planet at four different times as it approaches perihelion. It can be seen clearly that there is no evidence of condensation of the proto-planet during the early stages of its evolution, and it seems to be effectively torn apart by the Sun's field gradient before it reaches perihelion for the first time. The combined tidal action of the Sun and the receding star, both giving reinforcing 'pinch-type' forces towards the mean plane, almost reduces the planet model to two dimensions after a short time. Consequently, some of the individual mass point separations become very small, thus reducing the integration-step length to an unacceptably low value. To enable the computation to continue for an effective period of simulated time, the mass points of the upper and lower hemispheres were combined on to the 'equatorial' plane. The simulation was continued two-dimensionally until the centre of the model planet had completed its first orbit around the Sun.

From the above investigation it would seem that material captured by the Sun in the encounter described by Woolfson (1964), would not condense directly into planets due to self gravitational forces. It may be assumed, however, that the Sun would acquire an envelope of gas and dust of solar system dimensions from such an encounter, and it is possible that, in this envelope, planetary formation could take place by means of some accretion mechanism.

4. ALTERNATIVE STAR ENCOUNTERS

It was suggested in Section 2 that the chief obstacle to planetary condensation in the capture theory is the close approach to the Sun by the proto-planet shortly after it has been ejected from the star. In this section it will be shown that, as a result of a

closer, faster encounter, captured material may take up orbits in which it moves quickly away from the Sun soon after ejection and hence will have a good opportunity to condense.

The closest distance of safe approach by a star to the Sun during a ‘fast’ encounter, R , is given by (Jeans 1929)

$$R^2 = \frac{G^{1/2}M_{\odot}}{0.675v\rho_*^{1/2}} \tag{2}$$

where ρ_* is the density of the star, v is the velocity of the star and $M_{\odot}$ is the mass of the Sun. Suppose the star orbit is hyperbolic with eccentricity e , then

$$v^2 = G(M_{\odot} + M_*)(1 + e)/R$$

where M_* is the mass of the star. Substituting this in (2) gives

$$(R/r_*)^3 = 9.15M_{\odot}^2/M_*(M_{\odot} + M_*)(1 + e) \tag{3}$$

where r_* is the radius of the star. Values of R/r_* are given in Table I for a number of different ratios of star masses and two values of e .

From Table I it appears that the perihelion of the star orbit as described by Woolfson is approximately at the limit set by (3).

TABLE I
Least distance of safe approach by a star to Sun

$M_{\odot}/M_*$	R/r_*		$M_{\odot}/M_*$	R/r_*	
	$e = 1$	$e = 2$		$e = 1$	$e = 2$
1	1.32	1.15	7	3.04	2.66
2	1.83	1.60	8	3.19	2.78
3	2.18	1.90	9	3.33	2.91
4	2.44	2.13	10	3.46	3.02
5	2.67	2.33	15	4.02	3.51
6	2.86	2.50	20	4.43	4.87

To carry out a trial investigation of the types of orbit which captured material may take up as a result of an encounter between the Sun and another star, a simple three-body encounter was simulated numerically. The prospective planetary material was represented by a point of negligible mass situated at the boundary of the star which was itself also taken to be a point mass. Stars of differing masses were projected past the Sun on a number of orbits which were chosen on the basis of formula (3). The stars were taken to be in early stages of evolution and were considered to be collapsing and radiating away the gravitational energy so released. The equation of collapse can be written approximately

$$\frac{dr_*}{dt} = a \frac{L_* r_*^2}{GM_*^2}$$

where L_* is the luminosity of the star, and ‘ a ’ is a constant depending on the ratio of specific heats of the material of the star. Assuming an effective star temperature of 50°K , the velocity of collapse for a star of mass $0.25M_{\odot}$ with a radius $2.5 \times 10^{14} \text{ cm}$, was taken to be $2 \times 10^4 \text{ cm s}^{-1}$.

For a wide range of starting parameters it was found that the mass point at the boundary of the star was captured by the Sun; the results also confirmed the validity

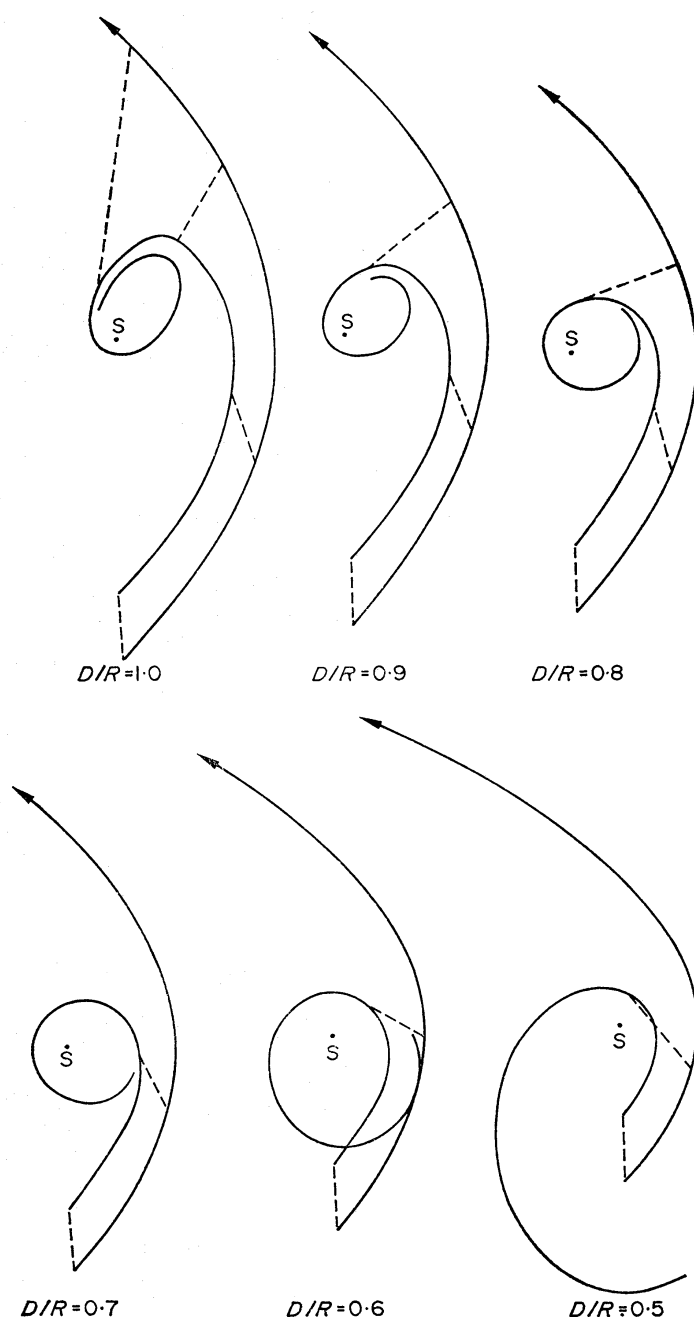


FIG. 5. Three body capture orbits. Star orbits are parabolic with various perihelia. $M_{\odot}/M_{*} = 4$, $r_{*} = 2.5 \times 10^{14}$ cm (see Table II). Dotted lines join simultaneous positions of the star centre and the escaping mass point.

of formula (3). It was noted that for large values of star orbit eccentricity ($e \sim 3$), if the perihelion was less than given by (3), the mass point was ejected from the star but it was not captured by the Sun. Table II shows some of the results obtained during this investigation and Fig. 5 shows the trajectories of the star and captured mass point in the encounters where $M_{\odot}/M_{*} = 4$ and $r_{*} = 2.5 \times 10^{14}$ cm.

In Table II D is the star perihelion distance, R the critical distance given by (3), r_{*} the star radius, ρ_{*} the star density, p the perihelion distance of the captured point, e its eccentricity, and H signifies that the orbit is hyperbolic.

It can be seen that in the closer encounters (Fig. 5), the captured mass point

TABLE II
Parameters of captured mass point in 3-body star encounter

(a) Parabolic star orbit									
$r_*(10^{14} \text{ cm})$		1.75		2.0		2.25		2.5	
$\rho_*(10^{-12} \text{ g cm}^{-3})$		11.1		7.46		5.24		3.82	
$\frac{M_\odot}{M_*} = 8$	D/R	$p(10^{14} \text{ cm})$	e	p	e	p	e	p	e
	1.0		H		H		H		H
	0.9	0.74	0.67	0.87	0.66	1.02	0.66	1.23	0.65
	0.8	0.92	0.52	1.10	0.50	1.25	0.50	1.40	0.49
	0.7	1.00	0.43	1.24	0.40	1.45	0.38	1.72	0.35
$\frac{M_\odot}{M_*} = 4$	0.6	1.12	0.34	1.43	0.30	1.73	0.28	2.05	0.26
	0.5	1.83	0.18	2.08	0.26	2.32	0.28	2.57	0.37
$\rho_*(10^{-12} \text{ g cm}^{-3})$		22.3		14.9		10.5		7.46	
$\frac{M_\odot}{M_*} = 4$	1.0	0.06	0.29	0.12	0.27	0.36	0.67	0.47	0.78
	0.9	0.09	0.26	0.34	0.56	0.51	0.66	0.59	0.67
	0.8	0.52	0.58	0.64	0.56	0.78	0.54	0.92	0.51
	0.7	0.43	0.56	0.53	0.58	0.69	0.55	0.82	0.53
	0.6	0.60	0.50	0.87	0.44	1.14	0.41	1.34	0.40
(b) Hyperbolic star orbit ($e = 2$)	0.5	2.10	0.50	2.42	0.55	2.68	0.56	2.89	0.57
	$\frac{M_\odot}{M_*} = 4$								H
	1.0								
	0.9							0.33	0.94
	0.8							0.53	0.84
$\frac{M_\odot}{M_*} = 4$	0.7							1.07	0.59
	0.6							1.40	0.62
	0.5							1.20	0.93

moves away from the Sun towards aphelion shortly after ejection from the star, whilst the orbits in the more distant encounters ($D/R > 0.6$) more closely resemble the orbit in Fig. 2.

The results obtained from the above 3-body encounters have been used to select initial parameters for the more complex numerical treatments of star encounters which are described in the next section.

5. A DETAILED STAR-ENCOUNTER MODEL

The star model chosen to represent a star in detailed encounter simulations was in many respects similar to the planet model described above. Since we are interested in the outer regions of the star from the point of view of its disruption, most of the mass (75 per cent) was concentrated in a central mass point. The remaining 25 per cent of the mass was arranged so that the mass points were situated on two concentric spherical surfaces. To achieve a fairly uniform distribution of points on the two surfaces, they were placed on the normals to the faces, and at the vertices of regular polyhedra. A hexoctahedron (3, 2, 1) was used for the outer shell, which was situated at the boundary of the star, and an icositetrahedron (3, 1, 1) formed the inner shell, which was situated at a distance in from the boundary equal to the mean separation of points on the two shells; permutation of the number triples (positive and negative values) gives the proportional direction cosines of the normals to the faces of these polyhedra. The total number of points is 124, but because of the symmetry of the configuration only 70 points need independent treatment.

One of the major difficulties in simulating the behaviour of such a star model, by numerical solution of the differential equations of motion of the individual mass points, is the necessity to compute the effect, on a single mass point, of all the other points in the model. If the 'planet model' method is used, and 70 points are treated independently, 2415 separate forces must be computed four times per integration step. When pairs of mass points approach each other very closely the simulation may be effectively halted since the integration step length must be shortened in the manner suggested in Section 3. This difficulty can to some extent be overcome by combining close pairs of points.

An alternative method of simulation is to fit some suitable geometrical surfaces to the configurations of points which were initially on spherical surfaces and to deduce the field of the star from the fields of bodies bounded by these surfaces. A suitable type of surface may be generated by the rotation of a 'limaçon' about its axis of symmetry (see Fig. 6). The volume enclosed by this surface will hereafter be referred to as a 'limacoid', and is similar to the shape a star may take up in the presence of a non-uniform gravitational field gradient. (c.f. Fig. 1, Woolfson 1964.) The equation of the limaçon in polar coordinates is given by

$$r = \frac{a}{k-1} (k \cos \theta - 1), \quad |\theta| \leq \arccos(1/k)$$

where a is the length of the axis and k is a measure of the shape of the curve (see Fig. 6). The gravitational field at the tip P of the limacoid of uniform density ρ_* may be determined analytically and can be written

$$F_P = \frac{\pi \rho_* G a (2k+1)(k-1)}{3k^2}.$$

It can be shown that the volume V and the position of the centre of mass $\bar{x}$ of the limacoid are given by

$$V = \frac{\pi a^3 (k-1)}{6k}$$

and

$$\bar{x} = \frac{\pi a^4 (k-1)(5k+1)}{60k^2 V}.$$

The gravitational field at any point Q (Fig. 6) due to the limacoid will be a function of R , ϕ , and k , and this has been computed by Gauss quadrature for a large number of positions for which $0 < R < a$, $0 < \phi < \pi$, and $1 < k < 6$. At each position, the centre of action, as a displacement from the centre of mass along the axis, and the ratio of the field to that of a sphere placed at the centre of action were computed. A polynomial surface of degrees 4, 9 and 4, in R , ϕ and k was fitted to each of the sets of data, and the coefficients were used in the encounter program to determine the gravitational field of the star at any point. During the encounter simulation the star is represented by two concentric limacoids plus an additional point mass at the star centre, and at the beginning of each integration step a limacoid is fitted to each shell of mass points.

The equations of motion of the mass points of the model star contained gravitational terms due to the Sun and the star, the latter being represented by the two limacoids plus the central mass point. A constant rate of collapse of the star was

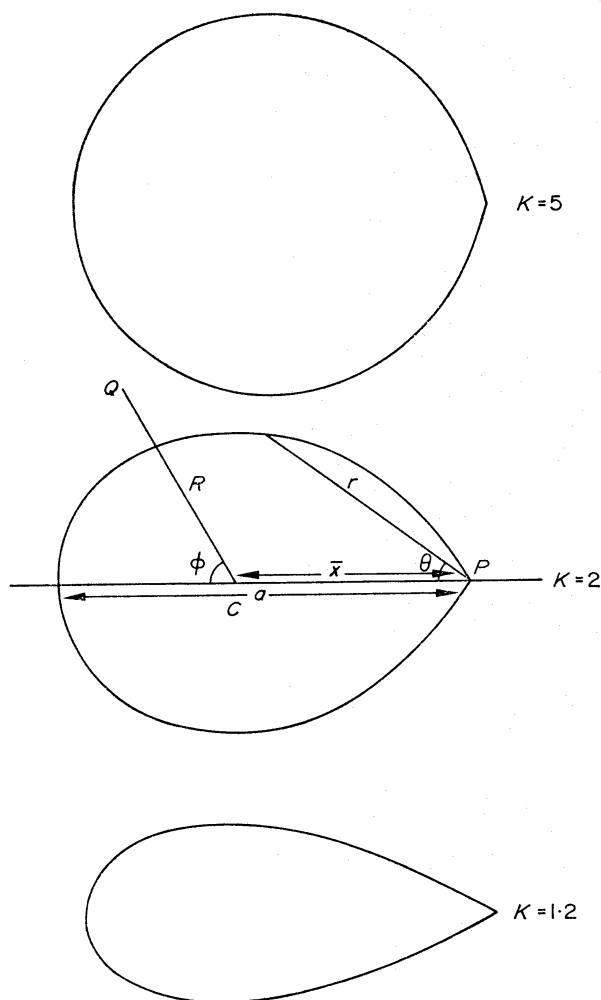


FIG. 6. The limaçon: $r = a(k \cos \theta - 1)/(k - 1)$ for three values of k .

maintained by adding a repulsion term to the normal expression for gravitational attraction. The modified gravitational term was written

$$F = \frac{GM_*}{R^2} (1 - v_c/V) \quad \text{if } v_c > 0$$

or

$$F = GM_*/R^2 \quad \text{if } v_c < 0$$

where M_* is the effective mass of the star, R is the distance to the centre of action of the star, V is the initial collapse velocity and v_c is the rate of collapse of the mass point. Mass points which were ejected from the star were not included in the limacoid fitting, and their equations of motion included gravitational forces due to other individual escaped points. Points whose distance from the star centre exceeds 1.4 times the original star radius are taken to be ejected.

Although the limacoid method should compensate to a large extent for the discrete nature of the model, which is simulating a continuous distribution of matter, it was felt necessary to introduce a further compensating factor—namely velocity sharing, which would inhibit localized mass point effects. These effects take place particularly in an escaping filament whose mass points do not contribute to the limacoid fitting. The sharing of velocities in the escaping filament tends to maintain

the point separations and also prevent 'pooling', without the introduction of repulsion forces, and has a general smoothing effect over the whole model star. The velocity of a point was taken as

$$v_i = \frac{\sum_{j=1}^n w_{ij} v_j}{\sum_{j=1}^n w_{ij}}$$

where

$$w_{ij} = 1 \quad \text{if } r_{ij} \leq r_m$$

$$w_{ij} = (r_m/r_{ij})^4 \quad \text{if } r_m < r_{ij} \leq r_c$$

and

$$w_{ij} = 0 \quad \text{if } r_{ij} > r_c$$

where r_{ij} is the distance between points i and j , and r_m, r_c are adjustable parameters. Sharing takes place at the end of every integration step. The degree of smoothing due to the velocity sharing will clearly depend on the critical distance r_c , and if this is too large it will seriously affect the encounter simulation. The actual value chosen ($r_c = 10^{14}$ cm) is a little greater than the initial mean point separation, ($\sim 9 \times 10^{13}$ cm) and so at the start of the simulation, only neighbouring points will have their velocities shared with each other. The value of r_m was taken as one tenth that of r_c .

The integration step length was recomputed at the end of every step using the method suggested in Section 3. Since at the start of the simulation the interactions of individual mass points are not considered, the step length does not depend on their separations and thus it can be much larger; the distance of the nearest point to the Sun is used to compute the step length. The separations of the ejected points have to be considered in the later stages of the simulation.

Pairs of mass points which approach each other closely, so that their mutually attractive force is greater than the disruptive force of the Sun, are combined at their centre of mass.

6. RESULTS FROM DETAILED STAR ENCOUNTER SIMULATIONS

Using the 'limaçon' method of star simulation as described above, it was found that the computation time required per Runge-Kutta integration step was about 20 per cent less than for a similar model in which all mass point interaction forces were computed. However, the time-saving will be greater if more complex models are used since the amount of computation required is directly proportional to the number of mass points and not to the square of the number as in the point interaction model. It is not easy to compare the accuracies of the two types of simulation, but since the 'limaçon' method eliminates, to a large extent, the localized point interactions, it is believed to be superior.

At the time of writing, four star encounter simulations have been completed using the 'limaçon' method. All four were of the 'fast' type, and their initial parameters are shown in Table III.

The results of the simulation of encounter 1 are shown pictorially in Fig. 7, and clearly show the development of a filament of fairly high density which is almost two-dimensional. The similarity of the configuration of mass points to the shape of a limaçon is quite marked in the early stages, but the approximation appears less good after the star's passage through perihelion. In this case the integration was continued

TABLE III

Parameters of encounters

Encounter	Star perihelion (10^{14} cm)	D/R	Eccentricity	Star radius (10^{14} cm)
1	3.1	0.5	1.01	2.5
2	3.1	0.6	1.01	2.25
3	2.8	0.5	1.50	2.5
4	3.0	0.6	2.00	2.5

Mass of star $M_* = 5 \times 10^{32}$ g = $\frac{1}{4}M_\odot$
 Single mass point $m = 1.01 \times 10^{30}$ g
 Radius of inner shell = 1.6×10^{14} cm
 Initial position of star $R = 3D$

Critical velocity sharing distance $r_c = 10^{14}$ cm
 Rate of collapse of star $-\dot{r} = 2 \times 10^4$ cm s $^{-1}$
 Initial rotation of star $\dot{\phi} = 10^{-10}$ rad s $^{-1}$
 Initial time step = 10^8 s.

for 100 steps (time = 1.5×10^9 s), requiring $3\frac{1}{2}$ hours computing time on the York ICL 4130 computer.

Subsequent behaviour of the filament was determined from a simpler computer model which involved the individual mass points of the filament but treated the remainder of the star as a point mass, and during these later stages of simulation many point combinations were made because of the numerous close encounters between the points of the relatively dense filament. The number of combinations was particularly large in the case of encounter 2 in which the initial star density was somewhat higher than in the other encounters. The complete set of orbital elements of the captured mass points in the four encounters are shown in Table IV.

The amount of material captured by the Sun in an encounter with another star will clearly depend to some extent on the parameters of the star orbit. We should expect that the faster, more hyperbolic orbits would result in less material being

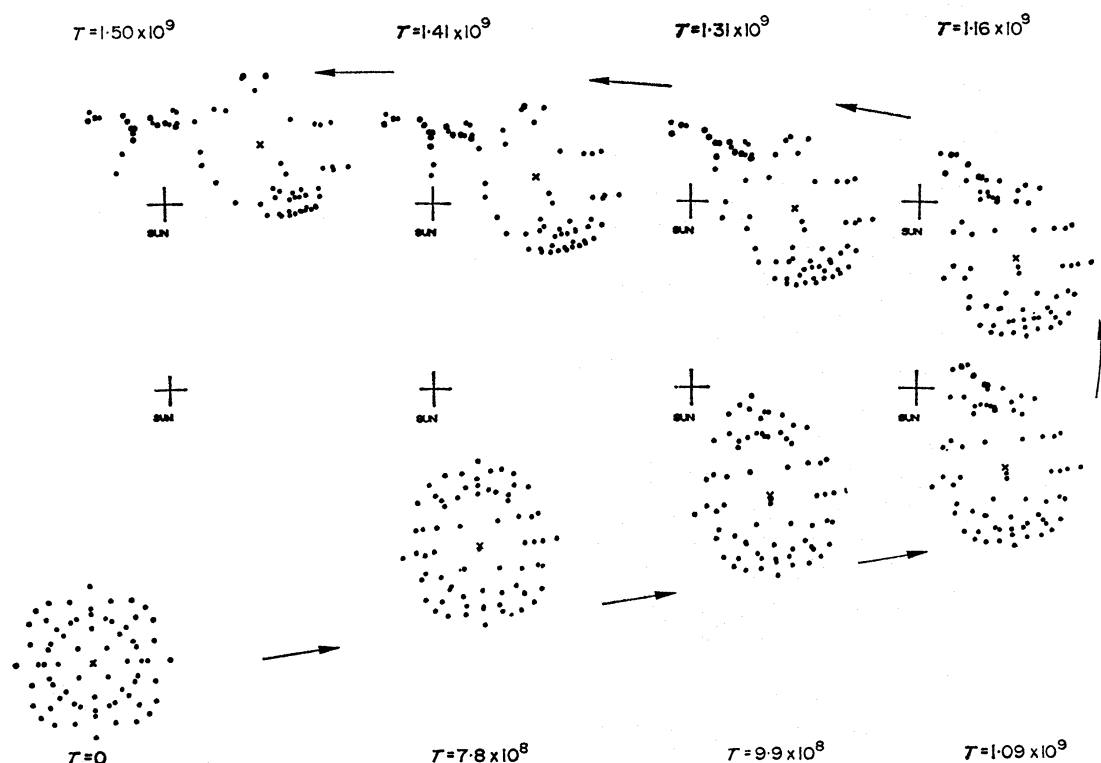


FIG. 7. Star encounter 1, showing development of the planetary filament.

TABLE IV
Orbits of captured material

Approximate time of ejection (10 ⁹ s)		Velocity components at time of ejection x y (10 ⁵ cm s ⁻¹)		Semi-major axis 'a', (10 ¹⁴ cm)	Eccentricity e	Perihelion $a(1 - e)$ (10 ¹⁴ cm)	Semi-latus rectum $a(1 - e^2)$ (10 ¹⁴ cm)	Mass of point (10 ³⁰ g)
Encounter 1								
1	1.17	-9.76	0.16	8.503	0.709	2.476	4.25	6.06
2	1.36	-7.54	-6.31	4.461	0.218	3.488	4.25	4.04
3	1.44	-10.97	-9.83	7.428	0.808	1.428	2.60	1.01
4	1.46	-6.08	-8.45	3.243	0.376	2.025	2.69	6.06
5	1.46	-5.96	-7.96	2.240	0.556	0.994	1.55	2.02
6	1.60	-3.71	-8.26	2.779	0.300	1.944	2.54	11.11
7	1.70	-2.40	-9.12	2.714	0.212	2.139	2.59	1.01
8	1.79	-1.28	-8.41	3.262	0.248	2.454	3.06	2.02
9	1.90	0.30	-7.29	1.937	0.592	0.790	1.23	1.01
10	1.92	-2.20	-5.05	4.326	0.776	0.968	1.73	1.01
11	2.00	-2.73	-3.64	4.258	0.783	0.923	1.66	1.01
12	2.20	3.49	-2.97	2.259	0.958	0.095	0.18	1.01
Encounter 2								
1	1.34	-7.36	-3.48	3.921	0.406	2.330	3.28	2.02
2	1.35	-8.13	-2.54	2.799	0.515	1.358	2.06	3.03
3	1.48	-7.18	-1.83	2.218	0.395	1.341	1.87	2.02
4	1.50	-6.03	-5.73	2.209	0.211	1.743	2.11	24.24
5	1.56	-10.53	-5.48	4.944	0.766	1.158	2.05	1.01
6	1.58	-10.37	-4.37	15.920	0.910	1.435	2.73	1.01
Encounter 3								
1	1.03	-10.06	-2.45	19.930	0.938	1.237	2.40	2.02
2	1.10	-11.36	-5.94	55.430	0.979	1.178	2.33	3.03
3	1.16	-7.30	-6.70	3.949	0.755	0.968	1.70	1.01
4	1.38	-2.74	-7.40	3.663	0.756	0.894	1.57	6.06
5	1.42	-2.60	-7.72	2.417	0.784	0.521	0.93	2.02
6	1.48	-5.31	-6.46	7.547	0.920	0.603	1.16	2.02
7	1.54	-3.47	-7.18	2.477	0.730	0.668	1.16	2.02
8	1.70	0.03	-7.58	2.517	0.780	0.554	0.98	5.05
9	2.10	2.37	-4.14	4.983	0.995	0.024	0.05	4.04
Encounter 4								
1	1.29	-0.97	-5.58	6.366	0.9939	0.039	0.077	2.02
2	1.34	-2.16	-4.63	11.230	0.9996	0.004	0.009	2.02
3	1.40	-0.07	-6.59	4.928	0.9608	0.193	0.379	7.07

N.B. $\dot{x}, \dot{y}$ are measured with respect to star centre.

captured than in the near parabolic cases, and the results in Table IV are consistent with this surmise. The least amount captured was in encounter 4 (orbital eccentricity 2), but in this case a similar amount of material ejected from the star had sufficient energy to escape from both stars. Some of the ejected material in encounter 3 also escaped from both stars but in encounters 1 and 2 the whole of the ejected filaments were captured by the Sun.

Although the orbital perihelia and semi-lati recta of the captured mass points fall within the range observed in the present-day solar system, the total mass captured by the Sun in each encounter, and the orbital eccentricities are considerably greater than contemporary values. These apparent difficulties will be discussed in a later section.

7. CONDENSATION OF PLANETS

Some of the problems associated with planetary condensation on the basis of the capture theory have been discussed in Section 2. Examination of the orbits of the captured material from encounter 1, described in Section 6, (Fig. 8) shows that the perihelia are all on the same side of the Sun as the star perihelion. Consequently

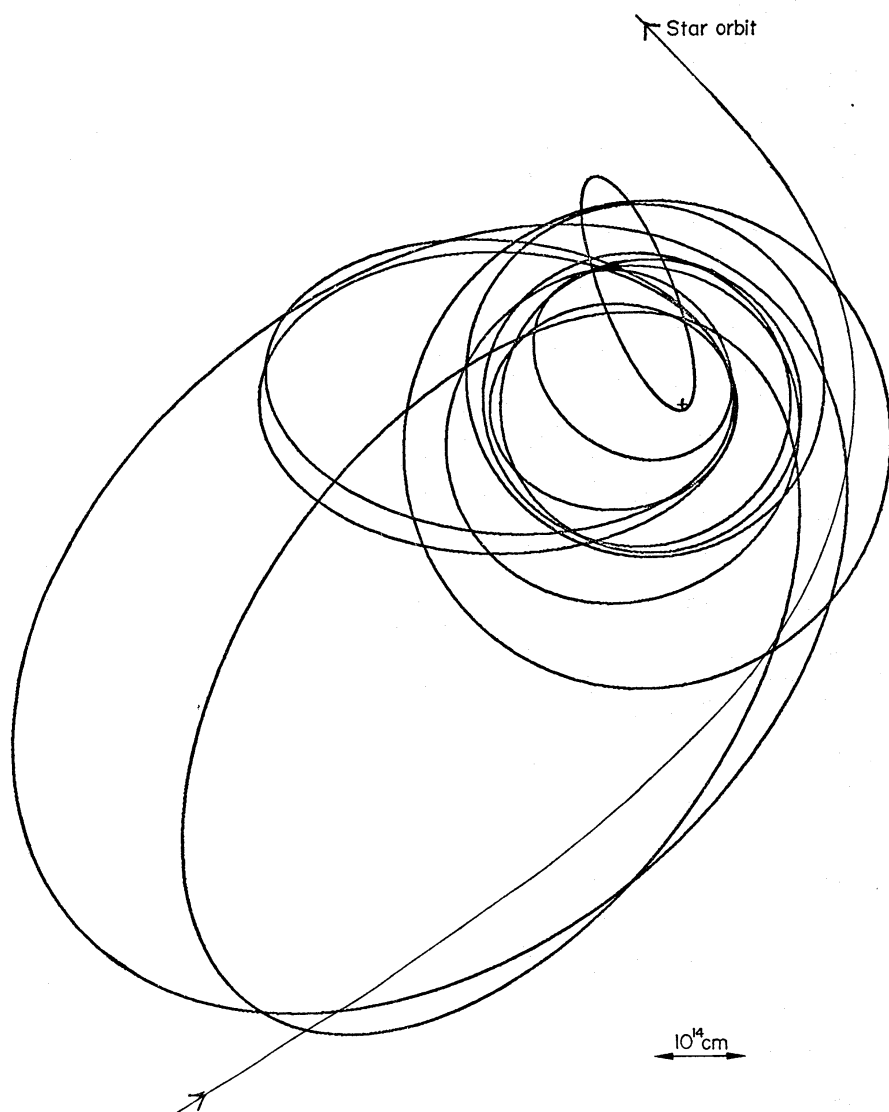


FIG. 8. *Star orbit and orbits of captured mass points from planetary filament in star encounter 1.*

many of the mass points in the escaping filament (Fig. 7) have moved past their perihelia at the end of the star simulation process and are therefore moving away from the Sun. Hence proto-planetary condensations will not have to withstand an increasing gravitational field gradient during the early stages of their evolution, as did the ill-fated proto-planet studied in Section 3. The density of the escaping filament is approximately $1.4 \times 10^{-11} \text{ g cm}^{-3}$, which is more than three times as great as that considered in the earlier work. It is clear that as a result of the relatively high density and decreasing field gradient, there will be a higher probability of planetary condensation after a 'fast' encounter than in those cases studied previously.

A similar analysis of the orbits of captured material in the other encounters listed above, indicates that favourable conditions exist for planet formation. In encounter 2 the filament density is even higher than in encounter 1, and although the densities are lower in the more hyperbolic encounters 3 and 4 the aphelia of the orbits produced are generally greater, thus giving the planetary material more time to condense before it re-approaches the Sun.

Planetary condensation in the captured filament from encounter 1 has been studied using a planet model simulation which is similar to that described in Section 3. Modifications made to the original planet model include:

- (i) Improved simulation of viscous forces;
- (ii) Redistribution of mass at intervals on a rectangular grid; and
- (iii) Revised integration-time step-length computation.

The viscosity simulation in the earlier planet model took no account of the changes in density of the proto-planet. A proto-planet, condensing under its own gravitation and the influence of the Sun's field gradient, suffers a general increase in density during the early stages, but later on a dense 'core' begins to develop in the centre whilst the outer regions become less dense, eventually to be removed by the tidal action of the Sun. The acceleration due to viscous forces acting on a planetary mass point was taken to be $\chi \rho v^2$ where χ is constant, ρ is a measure of the density of the planet neglecting the 'core', and v is the velocity of the point with respect to the centre point. The 'core' is represented in the simulation by the centre point of the planet, which as a result of point combinations eventually comprises a large fraction of the total mass of the planet. As before, the point combinations are made in order to avoid excessive reduction of the integration-time step.

The density distribution within the model planet was illustrated by output of a map at various stages of the simulation. This was produced by redistributing the mass of the planet on to a rectangular grid of points. It was assumed that each mass point of the model was situated at the mean $(\bar{x}, \bar{y})$ of a Normal distribution surface given by

$$f(x, y) = \frac{1}{2\pi\sigma^2} \exp \left[-\{(x-\bar{x})^2 + (y-\bar{y})^2\}/2\sigma^2 \right] \quad (4)$$

where σ^2 is the variance. The mass to be placed on each grid point was obtained by integrating (4) over the appropriate area. A map obtained by this method, on which equidensity contours have been drawn, is shown in Fig. 9.

By redistributing the masses at intervals during the simulation it should be possible to maintain large point separations and thus a large integration step. However, at the time of writing, uncertainties in the computation of velocity components of the new mass points makes this unacceptable.

The integration step-length computation was changed from that done for the original model because of the many mass point combinations which were found to take place. If we wish to integrate numerically the equations of motion of two mass points, it can be shown that the integration step-length should be given by

$$\Delta T \propto \left(\frac{r^3}{M} \right)^{1/2}$$

where r is the separation of the mass points, M is the combined mass of the points and ΔT is the time step. The initial value of ΔT was determined empirically, (for $M = 10^{29}$ g and $r = 1.66 \times 10^{13}$ cm, the best value of ΔT was found to be 4×10^7 s) and at the end of each integration step it was recomputed for each pair of

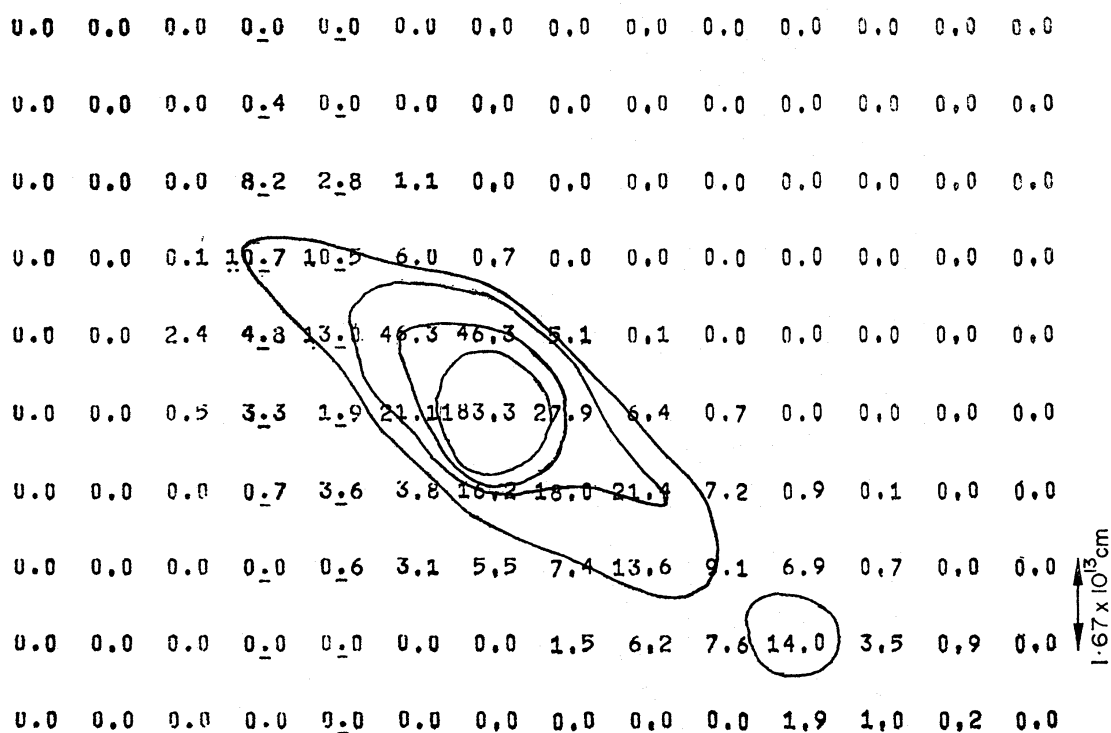


FIG. 9. Density map: planet 1 at 8.9×10^8 s from start of simulation. The masses of the points are in units of 10^{28} g and contours are drawn at 10, 20, 40 and 80 units, the minimum density within the '10' contour being 1.7×10^{-10} g cm $^{-3}$.

mass points, the smallest value being used as the overall integration step length. In the original planet model, M was not considered in the time step determination, but with the development of large mass points resulting from combinations in a condensing model planet, it can become significantly larger for some pairs than in the initial stages, and hence must be included in the computation.

At the time of writing, two proto-planets have been studied using the parameters of the ejected star filament from encounter 1. Both have shown signs of significant condensation.

At two positions, the mass points of the ejected star filament were replaced by spherical configurations of mass points representing the two proto-planets. The coordinates and velocity components of the two proto-planets were determined from the parameters of the filament mass points. Fig. 10 shows the filament mass points and the evolution of the two planets. As in the earlier planet experiments (Section 3),

TABLE V

Initial parameters of 2 proto-planets

Planet	Mass (10^{30} g)	Initial radius (10^{13} cm)	Initial density (10^{-11} g cm $^{-3}$)	Angular momentum (10^{48} g cm 2 s $^{-1}$)
1	6.0	5.0	1.15	-1.0
2	8.0	5.0	1.53	0.74

Number of mass points = 123

Initial point separation = 1.67×10^{13} cm

Critical separation for point combination = 10^{12} cm

Elbow parameter = 0.3

Viscosity parameter = 10^{-7} cm 2 g $^{-1}$.

the simulation was started in three dimensions but after a time the degree of flattening of the planet made it necessary to continue in two dimensions. The first density map on each orbit in Fig. 10 indicates the start of two-dimensional simulation.

The simulation was continued for each proto-planet until appreciable condensation had taken place. For planet 2, 1000 integration steps were made, requiring about 75 minutes computing on the ATLAS computer, and representing a time of 1.83×10^9 s. The time required for each integration step is proportional to the square of the number of mass points and so it is considerably lessened by the reduction of the number of points through combinations. At the end of the planet 2 simulation only 18 mass points remained.

Although both proto-planets showed evidence of appreciable condensation, large fractions of their initial masses were lost. This mass loss was particularly evident in

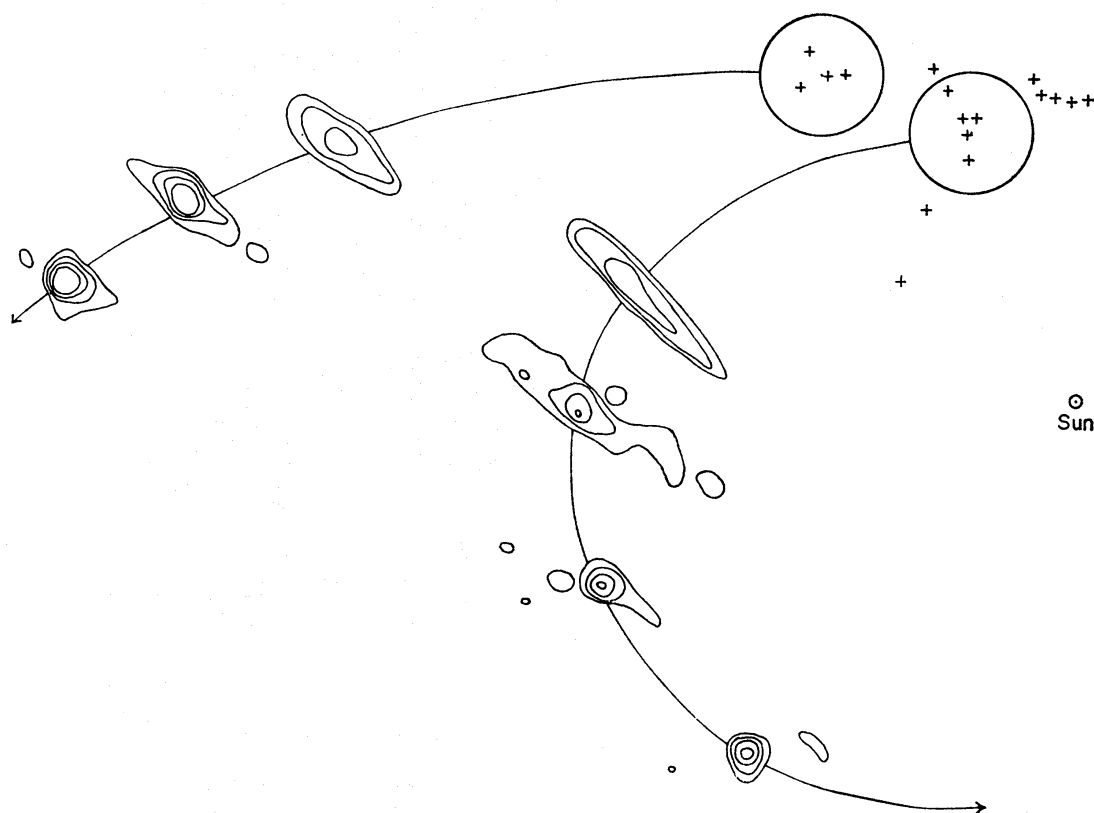


FIG. 10. The evolution of two proto-planets captured during star encounter 1. The relation is shown between the filament mass points (crosses) and the initial model planet configurations. Density map contours are drawn at 10, 20, 40 and 80 units.

TABLE VI

Final parameters of two proto-planets

Orbital elements	Planet 1	Planet 2
Semi-major axis (10^{14} cm)	11.91	3.017
Eccentricity	0.815	0.331
Perihelion	2.204	2.020
Semi-latus rectum	4.000	2.685
Mass (10^{30} g)	3.852	1.691
Density (10^{-11} g cm $^{-3}$)	3.326	1.460
Angular momentum (10^{46} g cm 2 s $^{-1}$)	7.727	8.377
	} within radius 1.67×10^{13} cm	
Mass (10^{30} g)	2.388	1.496
Density (10^{-7} g cm $^{-3}$)	8.907	5.58
Angular momentum (10^{46} g cm 2 s $^{-1}$)	5.272	-1.36
	} within radius 10^{12} cm	

the case of planet 2 which had a closer orbit than planet 1, and was therefore subjected to greater tidal forces.

In addition to the simulation of the two above planets, less complex simulations of captured material in closer orbits have been made. These suggest that large condensations could not be formed close to the Sun, but it seems likely that smaller condensations may be possible and this will be investigated at a later date.

A plausible theory of the origin of the planets ought to include the theory of the formation of planetary satellites. Unfortunately, the scale of the above described simulations is too large for a study of satellite formation since the orbits of the major satellites of the solar system have radii of the order of 10^{11} cm, whilst our model does not allow point separations of less than 10^{12} cm. It may be possible, however, to progressively reduce the scale of the simulation as condensation proceeds and hence extend it to include satellite formation. Despite the relatively large scale of the present planet model, it has been noted that as a result of the partial disruption of a proto-planet by the tidal action of the Sun, some mass points are left in elliptic orbits about the central condensation. Although this indicates the possibility of satellite formation, it is not claimed to be positive evidence and more complex simulation techniques will be required to verify the ejection of satellite material from a condensing planet.

A mechanism by which planetary satellites could be formed is suggested by our results. As a consequence of the non-uniform character of the gravitational field within which the planet condenses, it takes up a non-symmetrical shape and during the final stages of contraction a filament may be left behind in which satellites may form and remain in circular orbits in the plane of the planetary equator. The angular momentum of the central condensation formed in the planet 2 simulation suggests that it will begin to lose material from its edge when it has contracted to a radius of approximately 2×10^{11} cm. This radius is comparable to the orbital radius of Callisto (1.88×10^{11} cm), the outermost Galilean satellite of Jupiter.

8. UNCONDENSED MATERIAL AND THE PLANETARY ORBITS

It was suggested by Woolfson (1964) that the initially eccentric planetary orbits might be rounded off by the action of a resisting medium composed of gas and fine dust released during the star disruption; the final orbits would be circles of radii close to the original perihelia. Smart (1953) has investigated the effect of a static

resisting medium on a planetary orbit when the resistance per unit mass to the planetary motion is given by

$$R = cv/r^2$$

where c is a small constant, v is the velocity of the planet, and r is the distance of the planet from the Sun. In these circumstances it is found that the eccentricity and the major axis of the orbit are reduced by the medium. Kiang (1962) has verified this result and has further studied the evolution of direct orbits in a freely rotating medium. A freely rotating medium will have a braking effect on a planet when it is at perihelion, thus reducing the major axis of the orbit and the aphelion distance, but at aphelion the medium will tend to accelerate the planet and therefore increase its orbital major axis and perihelion distance. For a resistance law of the above type, the net effect will be to reduce both the major axis and eccentricity of the orbit. Kiang found that when the eccentricity has been reduced almost to zero, the radius of the orbit is about 70 per cent of the semi-major axis when the eccentricity was 0.5. This is a little less than the semi-latus rectum of an orbit of eccentricity 0.5.

There can be little doubt that a resisting medium would be formed from the debris resulting from the capture theory star encounter. It has been shown that in encounter 1 approximately $7\frac{1}{2}$ per cent of the mass of the star was captured by the Sun. This is about 14 times the mass of the present solar system, and therefore a great deal of mass is left to be accounted for. Clearly it cannot be supposed to have fallen into the Sun since its angular momentum must be far greater than the slowly rotating Sun. It may be shown that the angular momentum per unit mass of a uniformly dense rotating disc of radius r , is approximately

$$H = 0.8(GM_{\odot}r)^{1/2}.$$

If the outer radius, r , is 30 A.U., H is $1.96 \times 10^{20} \text{ cm}^2 \text{ s}^{-1}$ and since the mass is of the order of 10^{31} g , the total angular momentum would be greater than that of the planets and three orders of magnitude greater than that of the Sun. The major part of the resisting medium would consist of hydrogen gas and Hoyle (1960) has suggested that hydrogen would escape from the periphery of the solar system in a time scale of order 10^{14} s if the temperature of the gas is near the blackbody temperature (75°K) at this distance. Some of the angular momentum of the solid dust particles in the medium may be removed by the Poynting–Robertson effect. This material would be absorbed by the Sun when it was in circular orbit at the solar radius, with angular momentum of $3.1 \times 10^{18} \text{ cm}^2 \text{ s}^{-1}$ per unit mass. The absorption of the whole of the solid material of the resisting medium ($\sim 10^{29} \text{ g}$) would not be incompatible with the present-day slow rotation of the Sun. It is possible to make an estimate of the amount of material which could be absorbed in this way. The rate of absorption of energy by a blackbody of radius d at a distance r from the Sun is

$$\frac{dE}{dt} = \frac{L_{\odot} d^2}{4r^2} \quad (5)$$

where $L_{\odot}$ is the luminosity of the Sun. Now for a circular orbit the rate of decay due to the Poynting–Robertson effect is

$$\frac{dr}{dt} = -\frac{3L_{\odot}}{8rc^2\pi d\rho} \quad (6)$$

where c is the velocity of light and ρ is the density of the body. This may be combined with (5) to give

$$\frac{dE}{dr} = -\frac{mc^2}{2r}$$

where m is the mass of the body. Thus the energy per unit mass required by a body, for it to decay from a circular orbit radius R , to an orbit of solar radius $R_\odot$ is

$$E = \frac{c^2}{2} \log_e \frac{R}{R_\odot}.$$

If R is 30 A.U. then $E = 4 \times 10^{21}$ erg g^{-1} . In 10^9 years the Sun radiates 1.2×10^{50} erg and if we make the extreme assumption that all this energy is absorbed by the medium, we see that 3.3×10^{28} g may be removed in this way. This analysis applies only to the solid particles for which $d\rho > 5.7 \times 10^{-5}$ g cm^{-2} . For smaller particles, the Sun's radiation pressure exceeds its gravitational force and so these would tend to be driven out of the solar system.

Another mechanism which would reduce eccentricities of orbits has been suggested by Goldreich (1963). He considers the effect of tides, produced in a non-spinning satellite, which can only produce radial perturbation forces. The height of such a tide will oscillate as the distance from the primary changes due to the eccentricity of the satellite orbit. Energy will be dissipated in the satellite, but its angular momentum will remain constant, and so the eccentricity of the orbit will decrease whilst the semi-latus rectum will be unchanged. The final orbit will be a circle of radius equal to the original semi-latus rectum. It will be noted that this result is similar to that produced by a freely rotating resisting medium. From Table IV it can be seen that the semi-lati recta of the planetary orbits produced in encounter 1, vary between 4.25×10^{14} cm (28 A.U.) and 1.8×10^{13} cm (1.2 A.U.), (Neptune–Earth), which is very similar to the observed range of planetary orbits.

The distribution of mass of the captured material from the four star encounters in its evolved orbits, is compared with that of the solar system in Fig. 11. It can be

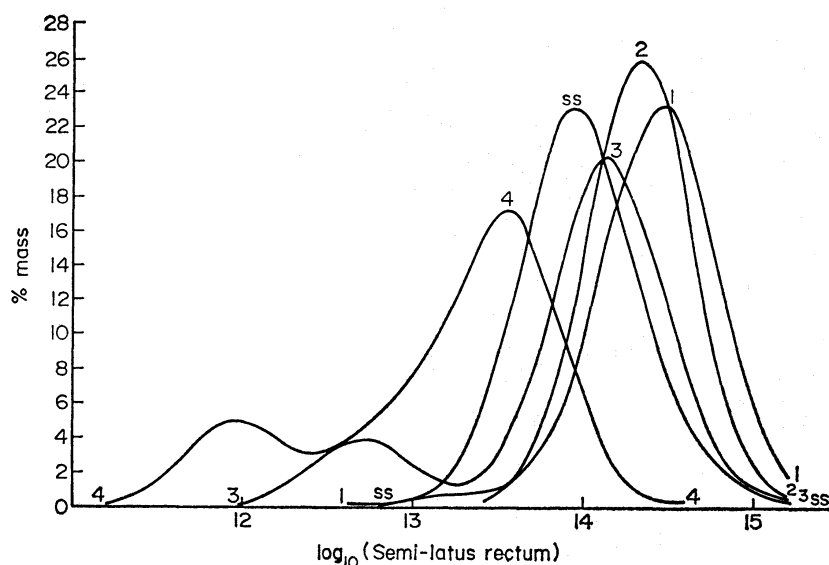


FIG. 11. Comparison of the solar system mass distribution with those of the captured material from four star encounters.

seen that all of the encounter distribution curves have well-defined maxima, although none of them coincide exactly with that of the solar system. The distinctive maxima are easily explained by reference to Fig. 12, in which the rate of ejection of material from the star (encounter 1) is plotted, showing a large maximum and thus indicating the ejection of a 'tapered' planetary filament which is much broader in the centre than at the ends. This figure also shows the variation of the semi-latus rectum of the captured material with time of ejection (c.f. Fig. 4, Woolfson 1964).

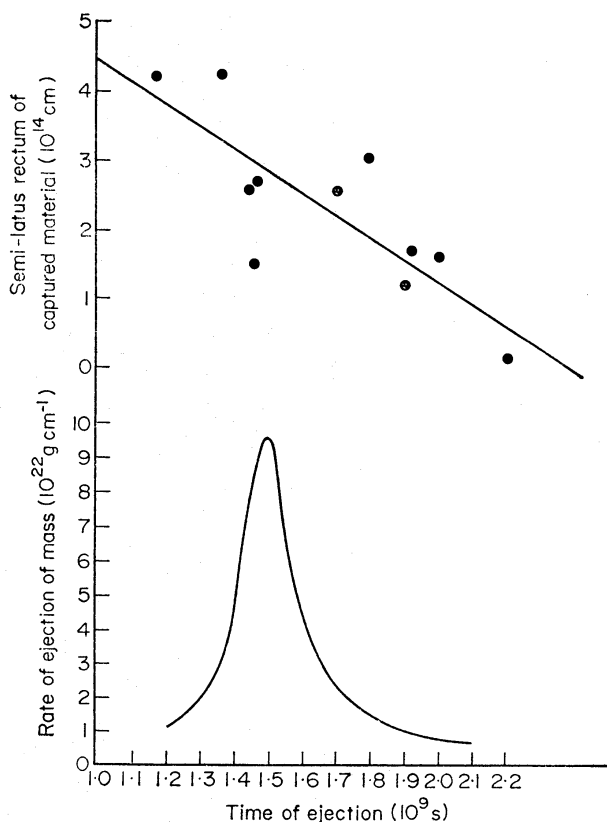


FIG. 12. Rate of ejection of mass by star and semi-latus recta of captured material (star encounter 1).

9. FREQUENCY OF PLANETARY SYSTEMS

Tidal theories of the origin of the solar system have sometimes been rejected on the grounds of low probability. It is proposed that the encounters described above could take place in a 'young' star cluster, in which the stars are comparatively close together. The probability of an encounter in such a cluster, between a relatively massive star like the Sun, and a lighter, more slowly evolving star, should be quite large. Although no accurate estimate of the probability of the formation of a planetary system on the basis of the 'capture' theory can be made, it is possible to calculate approximately the number of encounters which might occur in a dense star cluster. It can be shown that the number of encounters in unit time in a spherical star cluster is given by

$$N = 3n^2q^2v/4R_c^3$$

where R_c is the radius of the cluster, v is the mean velocity of the stars, q is the encounter parameter, and n is the number of stars. If we make allowance for the

gravitational force between a pair of interacting stars, the encounter parameter q can be estimated by

$$q^2 = a(a + 2GMv^{-2})$$

where a is the distance of closest approach and M is the mean star mass. Taking a cluster of 1000 stars, of radius 1 parsec, we find that there should be about 1 encounter in 10^5 years which satisfies the geometrical parameters of the 'capture' theory.

Aarseth (1968) has studied the evolution of star clusters using numerical techniques, and has verified the proposal that encounters may be common. If we make the assumption that stars are formed in large clusters in which encounters are frequent, then the 'capture' theory presents a plausible mechanism for the formation of planetary systems in large numbers throughout the galaxy.

10. CONCLUSIONS

The computations described above seem, on the whole, to reinforce the main aspects of the 'capture' theory; the one major difficulty in the original theory, concerning the condensation of the planetary material, has been resolved. The process of capture has been verified in a large number of encounter simulations, some of which were of much greater complexity than previously attempted. As before (Woolfson 1964), the process of capture has been found not to be sensitively dependent on the initial star parameters, and although the range of encounter orbits has been somewhat reduced by the gravitational field gradient requirements for planetary condensation, the initial parameters for a 'capture' encounter are not very restricted.

In the earlier study of the evolution of planets from the captured material (see Section 3), it was assumed that the density of the ejected filament would be approximately the same as that of the outer regions of the undisturbed star. It now seems that the filament density could be appreciably higher than this, and in the first star encounter (Fig. 7) it is about twice as high as the initial mean star density. Consequently, the likelihood of a planetary condensation surviving the effect of the Sun's gravitational field gradient is much greater than was thought previously. The planet simulations in Section 7 have demonstrated the possibility of the condensation of two planets, with masses approximately that of Jupiter, as a direct result of the capture of material by the Sun during an encounter with another star. The orbital perihelia (or semi-lati recta) of these bodies are not dissimilar to some of those observed in the present-day solar system. It is believed to be possible, by making simulations of the type described in this paper, to determine the initial parameters of a star encounter which would result in a planetary system very similar to our own. This would certainly be desirable, but a systematic variation of the parameters of the encounter program over a wide range of values would require more computer time than has been available to us.

The planetary evolution which has been simulated would apply to giant planets typified by Jupiter rather than terrestrial type planets. The scale of the simulations performed so far is too large for detailed application to the inner planets of the solar system. Experiments with a model planet in a closer orbit than those already described, (i.e. material from the end of the filament closest to the star) have suggested that a large condensation could not form close to the Sun, but the possibility that smaller condensations might occur cannot yet be ruled out.

In Section 7 the problem of satellite formation has been discussed qualitatively, and it will have been noted that the mechanism suggested for the formation of satellites is not the same as that for the planets, although it has often been assumed that the satellite systems are miniature replicas of the planetary system. Whilst this may be true of the dimensions of the satellite systems, it should be pointed out that their angular momenta are very small in comparison with those of their primaries. Since the reverse is true of the Sun-planets system, it is not unreasonable to assign different origins to the main system and sub-systems.

ACKNOWLEDGMENTS

Our gratitude is due to the Department of Computation of the University of York for their help in running a too-large problem on a too-small computer. We must also thank the Science Research Council for the use of the Chilton ATLAS, without which some of our more complex simulations would not have been possible.

Department of Physics, University of York, Heslington, York

Received in original form 1970 July 1

REFERENCES

- Aarseth, S. J., 1968. *Bull. Astr.*, **3**, 105.
 Goldreich, P., 1963. *Mon. Not. R. astr. Soc.*, **126**, 257.
 Hoyle, F., 1960. *Q. Jl R. astr. Soc.*, **1**, 1.
 Jeans, J. H., 1919. *Problems of cosmogony and stellar dynamics*, Cambridge University Press.
 Jeans, J. H., 1929. *Astronomy and cosmogony*, Cambridge University Press.
 Jeffreys, H., 1918. *Mon. Not. R. astr. Soc.*, **78**, 424.
 Kiang, K., 1962. *Mon. Not. R. astr. Soc.*, **123**, 359.
 Lyttleton, R. A., 1960. *Mon. Not. R. astr. Soc.*, **121**, 551.
 Rouskol, E. L., 1955. Les particules solides dans les astres, *Mem. Soc. R. Sci. Liege*, p. 650.
 Smart, W. M., 1953. *Celestial Mechanics*, Longmans.
 Spitzer, L., 1939. *Astrophys. J.*, **90**, 675.
 Woolfson, M. M., 1964. *Proc. R. Soc. A*, **282**, 485.